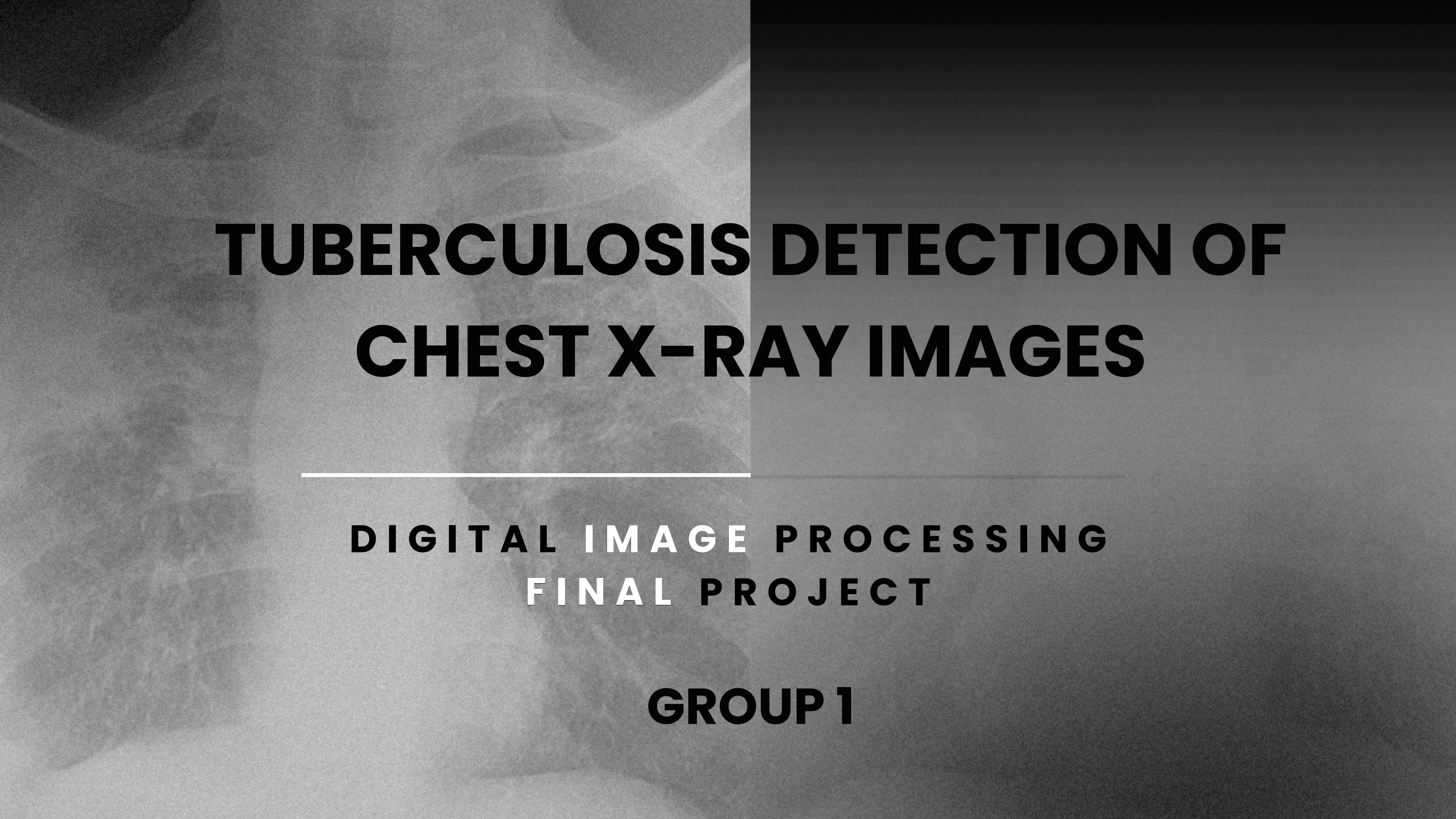




TUBERCULOSIS DETECTION OF CHEST X-RAY IMAGES

**DIGITAL IMAGE PROCESSING
FINAL PROJECT**

GROUP 1

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THE PROBLEM

Tuberculosis (TB) remains a leading cause of death worldwide, especially in low and middle income countries where access to advanced hospital tools is limited. Chest X-rays are a widely available, low cost screening method, but accurate interpretation often requires experienced radiologists. While some approaches use deep learning to automate TB detection, it may not be feasible in all environments.

This project explores a digital image processing approach, offering a lightweight, interpretable, and resource efficient option for screening in resource-limited environments.



OUR SOLUTION

Our project introduces an alternative, feature-based machine learning pipeline that is computationally efficient and provides interpretable results. The methodology follows a structured, multi-stage process. First, the raw chest X-ray images go through a pre-processing step using Contrast Limited Adaptive Histogram (CLAHE) and median filtering to improve contrast and reduce noise. After the images are enhanced, the system automatically segments the lung regions. From these segmented lungs, we extract several numerical features related to geometry, intensity, and texture. These features are then used to train a Random Forest classifier that can distinguish between Normal and Tuberculous cases.

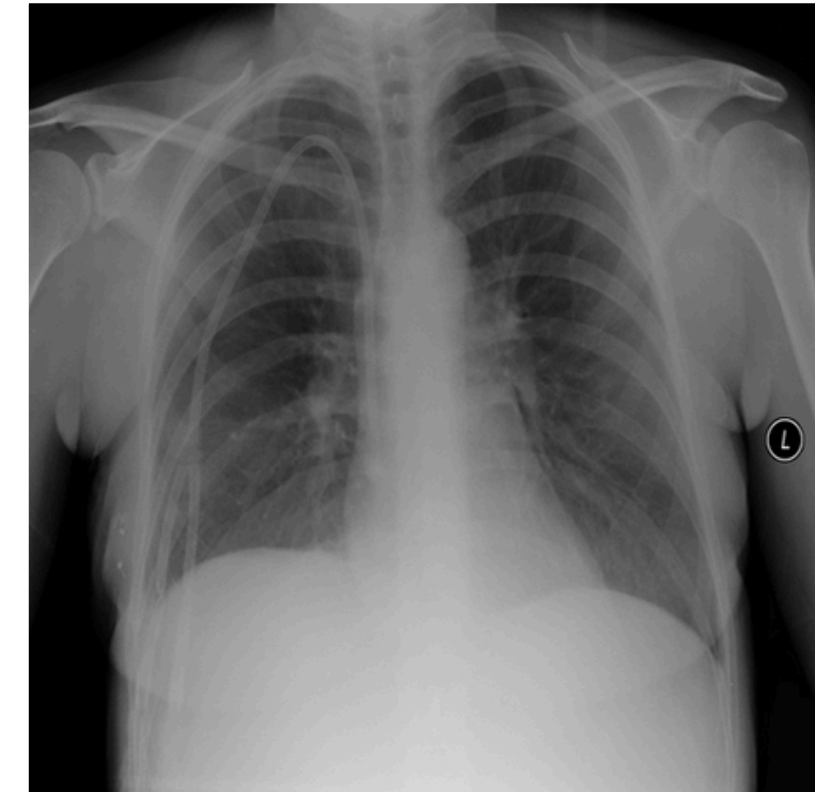


DATASETS

Our project used the public “Tuberculosis (TB) Chest X-ray Dataset” available on Kaggle [1]. The posterior - anterior chest radiographs in this dataset have been carefully selected for the purpose of classifying tuberculosis. There are 4.200 images in all, divided into two different classes:

Normal: 3,500 images of healthy individuals.

Tuberculosis: 700 images showing radiological evidence of tuberculosis



Normal Image from the provided datasets [1].



Tuberculosis Image from the provided datasets [1].

MODEL RESEARCH AND SELECTON

After extracting visual features from digital image processing, various machine learning models were evaluated to determine the most suitable one.

We use Random Forest Classifier:

- High Accuracy & Overfitting Resistance: Uses an ensemble fo many decison trees, reducing reliance on a single model and improving generalization.
- No Need Feature Scaling: Handles features with very different numerical ranges without preprocessing.
- Bulit-in Feature Importance: Identifies which extracted feature contribute most to TB detection, improving interpretability,
- Captures Complex Non-Linear Patterns: Tree-based structure models “if-then” relationships typical in medical data, making it suitable for detecting subtle TB indicators.

CHALLENGES/ LIMITATIONS

- **File and Path Management (The 0 samples Error)**

Problem: Running the “Main Loop” processed 0 samples because the code couldn’t find any images.

Mistake: NORMAL_PATH and TB_PATH pointed only to folders, not the images.

Solution: Added a wildcard (`/*.png`) so glob detects all .png files inside the folders.

- **Segmentation Issue (Only One Lung Detected)**

Problem: `segment_lungs_v2` and `extract_lung_features` worked on the test image but skipped most of the dataset images (“Did not find exactly two lungs”).

Mistake: Thresholds and size filters were overfitted to one test image and not general for 4000+ images.

Solution: Modified `extract_lung_features` by changing the `min_size` filter to a percentage of the largest object (`np.max(sizes) * 0.2`), making the process adaptive and increasing processed samples from almost 0 to ~2700.

CHALLENGES/ LIMITATIONS

- **Segmentation Function Not Working on Dataset**

Problem: The `segment_lungs_v2` and `extract_lung_features` functions worked perfectly on our testing image, but on the image dataset it skipped the majority of the image, giving the message “Did not find exactly two lungs.”

Mistake: The settings were overfitted to the single test image, which could have different contrasts, noises and sizes.

Solution: We changed the `min_size` filter from a hard-coded value to be a percentage of the largest object ($\text{np.max}(\text{sizes}) * 0.2$). This allowed it to correctly find lungs of different sizes and images, which took processed sample count from nearly 0–2,700.

- **Black Outline of the Image Disrupting Bounding Boxes**

Problem: Bounding boxes captured two halves of the entire image instead of the lungs.

Mistake: Thick black border wasn't removed, so the bounding box algorithm took the full image.

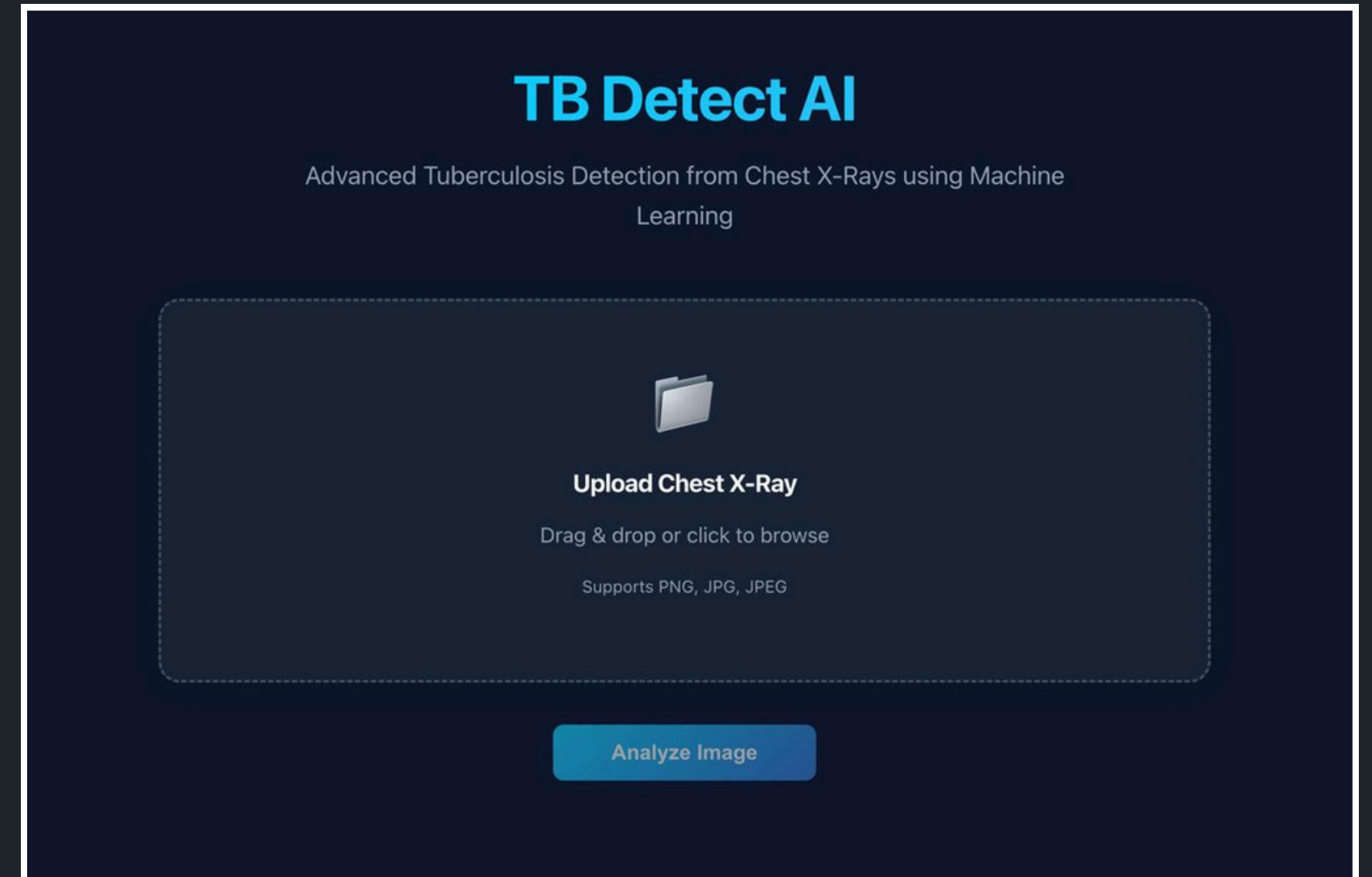
Solution: Cut out the black border so bounding boxes correctly target the lung regions.

FRONT-END/ SYSTEM IMPLEMENTATION

A front-end interface was developed to improve usability and move beyond a notebook-based prototype. The interface enables an end-to-end workflow (upload->analyze->result) so non-technical users can run inference without interacting with code.

The results are shown in a clear and responsible format, such as predicted class, confidence score, and extracted feature summaries. These support model interpretability and transparency,

Overall, the front-end demonstrates deployment readiness and allows the model to be tested in realistic user-interaction conditions as a decision support tool, not a definitive diagnostic system.



RESULTS

- **Dataset Assembly and Preparation**

For each X-ray with two successful segmented lungs, a single feature vector was created. Each vector contained 5 features for the left lung (area, perimeter, mean intensity, standard deviation, 5 features for the right lung, 3 additional difference features (area_diff, entropy_diff, intensity_diff) to capture left-right asymmetry. From the original 4,200 images 3,383 images were successfully processed into structured feature vectors. The final dataset was split into 2,706 training samples and 677 testing samples. The model was trained only on the training set reserved for final evaluation

- **Model Training**

A Random Forest Classifier, an ensemble learning method used due to its high performance and robustness on tabular data, was selected for the classification task. The model was configured with 100 individual decision trees and trained the feature vectors from the 2,076 samples.

RESULTS

- Performance Evaluation

The model achieved a strong overall accuracy of **92.76%**. The true negatives where 564 were actually normal and the model predicted it correctly. The “64” m images were actually “Tuberculosis” and the model predicted it correctly. Then the “13” shows false positives where it’s actually normal but the model prediction is wrong. The “36” are “Tuberculosis” but the model predicted wrong. So, it is leads to 628 correct predictions out of 677 giving 92.76% accuracy.

--- Model Evaluation ---				
Overall Accuracy: 92.76%				
--- Classification Report ---				
	precision	recall	f1-score	support
Normal (0)	0.94	0.98	0.96	577
Tuberculosis (1)	0.83	0.64	0.72	100
accuracy			0.93	677
macro avg	0.89	0.81	0.84	677
weighted avg	0.92	0.93	0.92	677
--- Confusion Matrix ---				
[[564 13]				
[36 64]]				

CONCLUSION

This project successfully presents the design, implementation, and evaluation of a system for the automatic detection of tuberculosis in chest X-ray images.

Model Performance

- Overall accuracy: 92.76%
- Strong performance on the Normal class (98% recall), making it effective for ruling out negative cases.
- Lower recall for the Tuberculosis class (64%), resulting in significant false negatives an important clinical limitation.

• Key Observations

- Segmentation was the most sensitive stage, small changes significantly affected the number of usable images.
- Hand-crafted features (entropy, intensity, asymmetry) improved interpretability compared to deep learning.

• Future Improvements

- Use class-weighted or cost-sensitive training to reduce false negatives.
- Add richer features (e.g., Haralick textures, multi-scale gradients).
- Explore hybrid pipelines combining feature extraction with lightweight neural models.
- Improve segmentation robustness with more adaptive methods.

A classical image-processing + feature-based ML pipeline offers an interpretable, efficient solution for TB screening, with strong potential for large-scale, low-cost prescreening especially if future work reduces false negatives and strengthens segmentation reliability.



THANK YOU
